PACKAGE MANAGEMENT

==========

1. It is a concept which can be used to manage 1st party softwares.
2. 1st party software means the software is developed by the OS vendor and available with the OS DVD.
3. In this section we will learn how to install,uninstall,update,refresh and collect information about a package.
4. To perform package management we need some package managers.
5. Example :
    a. RPM : Redhat Package Manager
    b. YUM : Yellowdog Updater Modified
    c. DNF : Dandified Yum
    d. APT
    e. DPKG
6. Using RPM,YUM and dnf we can manage packages in RHEL,CentOS,Fedora and Amazon Linux
7. Using APT we can manage packages in Debian Based OS [ubuntu,Kali]

What is a package ?

- It is a setup or collection of programs.

What is Dependency ?

- These are smaller or minor packages which are required to be installed in order to use the major package.

RPM :-

=========

1. To find out all packages installed in a system

```
[root@localhost ~]# rpm -qa | less
```

RPM : name of the package manager

q : query

a : all

| less : to activate scrolling

2. To get details of a package

Package name with details :- gzip-1.12-1.el9.x86_64

gzip : package name

1.12 : release information

1 : version of the package

el9 : enterprise linux 9

X86_64 : architecture of the package [x86_64 : 64 bit and i86 : 32 bit]

3. To collect information about a specific package / to find out a package is installed or not :

```
[root@localhost ~]# rpm -qa | grep gzip
gzip-1.12-1.el9.x86_64
[root@localhost ~]#
```

4. To get information about a package

```
[root@localhost ~]# rpm -qi gzip
Name        : gzip
Version     : 1.12
Release     : 1.el9
Architecture: x86_64
Install Date: Sat 21 Dec 2024 09:27:10 AM IST
Group       : Unspecified
Size        : 377013
License     : GPLv3+ and GFDL
Signature   : RSA/SHA256, Fri 22 Apr 2022 01:46:33 PM IST, Key ID 199e2f91fd431d51
Source RPM  : gzip-1.12-1.el9.src.rpm
Build Date  : Fri 22 Apr 2022 01:13:54 AM IST
Build Host  : x86-vm-54.build.eng.bos.redhat.com
Packager    : Red Hat, Inc. <http://bugzilla.redhat.com/bugzilla>
Vendor      : Red Hat, Inc.
URL         : http://www.gzip.org/
Summary     : The GNU data compression program
Description :
The gzip package contains the popular GNU gzip data compression
program. Gzipped files have a .gz extension.

Gzip should be installed on your system, because it is a
very commonly used data compression program.
[root@localhost ~]#
```

5. To install a package using RPM :

i : Install v : verbose h : print output U : upgrade F : refresh q: query

If there is a package installed on the system named “x.1.0”

Now a new version of the package is released named “x.1.1”

1. For installing if we use “i” then the system will keep both versions of the package.
2. Now if we use “U” for installing the system will check if there is any older version of the package available or not in the system. If available then it will replace the older version with the newer version.
3. Now if we use “F” it will also check if there is any older version of the package available in the system , if an older version is found , it will upgrade the older version , if not found it will not install the new package , it will cancel the installation.

Installing a Package using RPM :

==========

We want to install ftp service :

1. In linux ftp is known as “vsftpd” : very secure FTP Daemon

```
[root@srv1 ~]# rpm -ivh /media/AppStream/Packages/vsftpd-3.0.3-49.el9.x86_64.rpm
Verifying...                          ################################# [100%]
Preparing...                          ################################# [100%]
Updating / installing...
   1:vsftpd-3.0.3-49.el9              ################################# [100%]
[root@srv1 ~]#
```

2. To verify the package is installed or not

```
[root@srv1 ~]# rpm -qa | grep vsftpd
vsftpd-3.0.3-49.el9.x86_64
[root@srv1 ~]#
```

Uninstalling a package :

==========

```
[root@srv1 ~]# rpm -e vsftpd
[root@srv1 ~]#
[root@srv1 ~]#
[root@srv1 ~]# rpm -qa | grep vsftpd
[root@srv1 ~]#
```

How to find dependencies for a major Package

==========

```
[root@srv1 ~]# rpm -ivh /media/AppStream/Packages/httpd-2.4.53-7.el9.x86_64.rpm
error: Failed dependencies:
        httpd-core = 0:2.4.53-7.el9 is needed by httpd-2.4.53-7.el9.x86_64
        system-logos-httpd is needed by httpd-2.4.53-7.el9.x86_64
[root@srv1 ~]#
```

YUM : Yellowdog Updater Modified

==========

1. It is a package management system.

2. It is used to install 1st party softwares.
3. Yum is used to find dependency packages for a major package.
4. It can find out the dependencies and also it will install them.
5. Current version of YUM is known as DNF
6. DNF : Dandified YUM
7. Commands are the same as yum in DNF but the internal API (Application Programming Interface) is different , for that dnf performs better in installing and searching packages.

Difference between YUM & DNF :

Feature	YUM	DNF
Performance	Slower, especially during large updates.	Faster, more memory efficient, and better at resolving dependencies.
Dependency Resolution	Uses a public API-based method.	Uses the advanced `libsolv` library for faster and more accurate results.
Development	Primarily developed in Python.	Developed in C and Python, with a more stable and well-documented API.
Features	Limited plugin architecture.	Improved plugin architecture, supports modularity, and has automatic bug reporting.
Command Syntax	`yum`	`dnf` (though `yum` is often aliased to `dnf` for backward compatibility).
Compatibility	More compatible with older systems.	Designed for modern systems, but is the default on newer distributions that previously used YUM.

YUM Models

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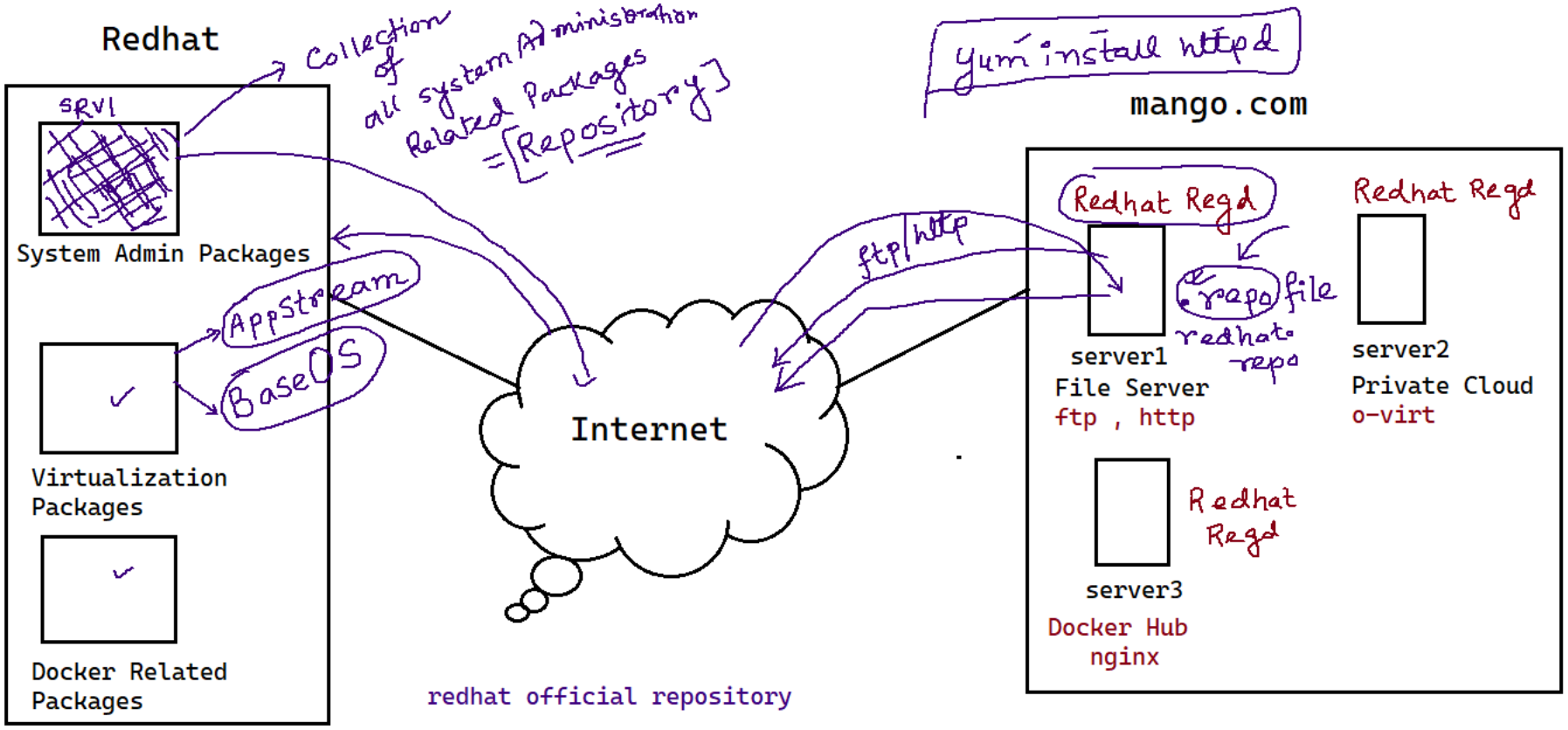
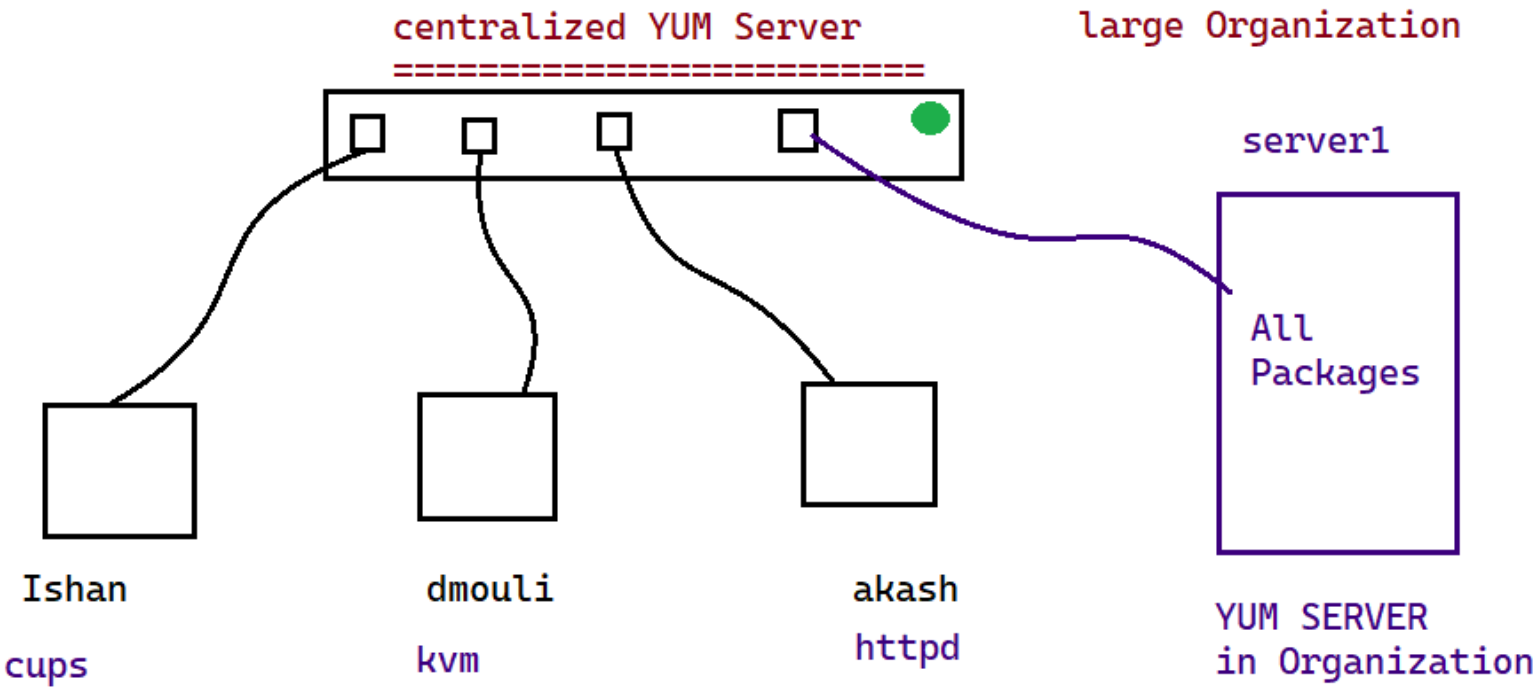
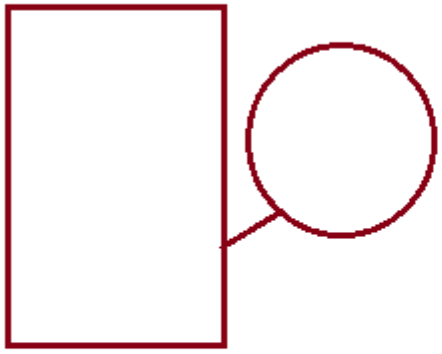
1. Yum server in Redhat Official Repositories
2. Centralized yum server hosted in company

3. Decentralized yum installed in a host system




centralized YUM Server
==========
large Organization
server1
All
Packages
Ishan
cups
dmouli
kvm
akash
httpd
YUM SERVER
in Organization

decentralized YUM



Standalone System

Preferred in LAb / Testing ENvironment

BaseOS:-
==========

a. The BaseOS repository contains the core set of operating system functionality that provides the foundation for all installations.
b. This content is available in the form of RPM packages
c. It supports all previous releases of Red Hat Enterprise Linux.

AppStream
==========

a. The AppStream repository contains additional user-space applications, runtime languages, and databases in support of the varied workloads and use cases.
b. Content in AppStream is available in one of two formats - the RPM packages and an extension to the RPM format called *modules*.

STEPS TO CONFIGURE FTP SERVER FOR CENTRALIZED YUM SERVER
==========
Server1 :

1. Set IP Address using nmtui
2. Disable firewalld
    a. # systemctl stop firewalld
    b. # systemctl disable firewalld
3. Connect and mount the dvd
    a. # mount /dev/sr0 /media
4. Install ftp service using rpm
    a. # rpm -ivh /media/AppStream/Packages/vsftpd<tab>

5. Restart & enable ftp service
    a. # systemctl enable --now vsftpd
6. When we install ftp service a automatic storage is allocated to ftp service
   Location :- cd /var/ftp/pub
7. Create a folder inside /var/ftp/pub
    a. # mkdir /var/ftp/pub/iso

8. Now copy all content from /media to /var/ftp/pub/iso
      # cp -rv /media/* /var/ftp/pub/iso

9. Now allow ftp for anonymous users
   Open configuration file of ftp
      # vim /etc/vsftpd/vsftpd.conf
      Change anonymous_access=YES
      Save and exit

10. Restart the ftp service
    a. # systemctl enable --now vsftpd

Client end

==========

1. Set IP Address
2. Turn off firewall
3. Check connectivity with the server using ping
4. Now open /etc/yum.repos.d location

   # cd /etc/yum.repos.d

5. Create a .repo file

   a. vim local.repo

      Enter the following content

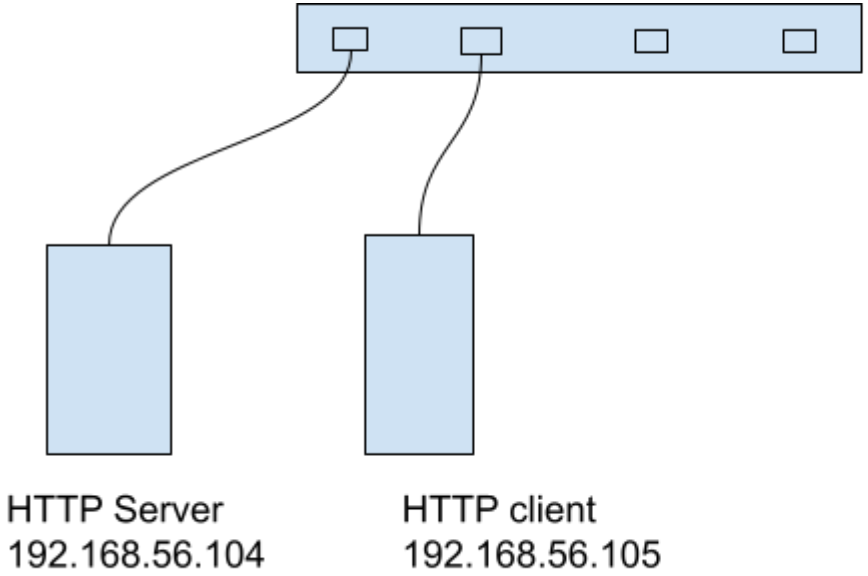
```
[appstream]
name=appstream repo
baseurl=ftp://192.168.10.1/pub/iso/AppStream
enabled=1
gpgcheck=0
[baseos]
name=baseos repo
baseurl=ftp://192.168.10.1/pub/iso/BaseOS
enabled=1
gpgcheck=0
```

Save and exit

6. Refresh and update yum entry
    a. # yum clean all
    b. # yum update

7. Install packages from yum server
    # yum install http* -y

HTTP BASED YUM SERVER
==========



Server End :

==========

1. Install http Package [Either from a FTP Server / Register The system/Configure Local YUM Server]

```
[root@client1 ~]# yum install httpd -y
```

2. Restart and enable the http service

```
[root@client1 ~]# systemctl enable --now httpd
[root@client1 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2024-12-31 10:02:20 IST; 9min ago
```

3. Connect DVD , then mount and copy all packages to the http default location

```
[root@client1 ~]# mount /dev/sr0 /media
```

Create a folder inside /var/www/html named allpackages

```
[root@client1 ~]# cp -rv /media/* /var/www/html/allpackages/
```

4. Again restart the http service

```
[root@client1 ~]# systemctl enable --now httpd
[root@client1 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2024-12-31 10:02:20 IST; 9min ago
```

Client End

==========

1. Check connectivity with the server using ping
2. Create a .repo file inside /etc/yum.repos.d

```
[root@client2 ~]# cd /etc/yum.repos.d/
[root@client2 yum.repos.d]# ls
[root@client2 yum.repos.d]#
[root@client2 yum.repos.d]# vi local.repo
```

3. Now enter details inside the file regarding the http yum server

```
[appstream]
name=appstreamrepo
baseurl=http://192.168.56.104/allpackages/AppStream
enabled=1
gpgcheck=0
[baseos]
name=baseosrepo
baseurl=http://192.168.56.104/allpackages/BaseOS
enabled=1
gpgcheck=0
[root@client2 ~]#
```

4. Clean and update the yum client

```
[root@client2 ~]# yum clean all
```

```
[root@client2 ~]# yum update
```

5. For testing install a package from the yum server

```
[root@client2 ~]# yum install vim -y
```

Local Yum Server / Decentralized YUM Server

==========

1. Only requirement is the ISO image of the OS
2. Connect the DVD
3. Mount the dvd

```
[root@client1 ~]# mount /dev/sr0 /media
```

4. Create a .repo file inside /etc/yum.repos.d named local.repo
5. Enter the following content

```
[appstream]
name=local1
baseurl=file:///media/AppStream
enabled=1
gpgcheck=0
[baseos]
name=local2
baseurl=file:///media/BaseOS
enabled=1
gpgcheck=0
[root@client2 ~]#
```

6. Save and exit
7. Now update yum

```
[root@client2 ~]# yum update
```

8. For testing install any package

```
[root@client2 ~]# yum install vim -y
```